

Proposal: Chatbot Roadmap Update (Week 5 Onwards)

Purpose

This proposal recommends updating the DiscountMate chatbot implementation from Week 5 onwards. The original roadmap uses a traditional NLP pipeline. The revised design adopts an LLM agent with Model Context Protocol (MCP), LangGraph orchestration and optional LangFlow prototyping while keeping the same business objectives.

Current Architecture (Before)

User → Intent Classification → Entity Extraction → Product Matching → Database Query → JSON Response.

This design requires separate intent models, entity extraction logic and routing rules. As more features are added, additional pipelines and maintenance effort are required.

Proposed Architecture (After)

User → LLM Agent → Tool Selection → MCP Tools → Database/API → JSON Response.

The LLM interprets requests and selects the correct tool. MCP abstracts backend services while LangGraph coordinates multi-step workflows and conversation state.

Before vs After Comparison

Area	Before	After
Reasoning	Intent classifier + rules	LLM agent with tool calling
Database Access	Direct SQL/API integration	MCP tools abstract backend
Workflow	Linear pipeline	Graph-based orchestration (LangGraph)
Context Awareness	Little or none	Current screen and session context
Scalability	New intent = new pipeline	New capability = new MCP tool
Maintenance	Multiple NLP components	Modular tools and workflows
Future Expansion	Requires architecture changes	Add tools without redesign

Representative User Stories

US-01 Price Comparison

As a shopper, I want to ask 'Where is Coke cheapest?' so that I can quickly find the lowest price.

US-02 Context Awareness

As a shopper viewing a product, I want to ask 'Is this cheaper anywhere else?' so the chatbot automatically understands the current product without repeating its name.

US-03 Product Search

As a shopper, I want to search using natural language such as 'Show me 2L milk under \$5' instead

of manually applying filters.

US-04 Product Substitution

As a shopper, I want cheaper alternatives to a selected product so I can reduce my shopping cost.

Why the Updated Structure is Better

Industry Alignment

Modern LLM applications increasingly separate reasoning from external actions using tool calling and protocols such as MCP, rather than embedding business logic directly into prompts.

Modularity

Each capability becomes an independent tool, reducing coupling between AI and backend systems.

Maintainability

Backend schema changes are isolated behind MCP interfaces, reducing impact on chatbot logic.

Context Awareness

Using application context enables shorter, more natural conversations and reduces repeated user input.

Extensibility

Future capabilities such as recommendations, shopping lists or retailer integrations can be added by introducing new tools instead of redesigning the chatbot.

Evidence Supporting the Change

- MCP is an emerging open protocol adopted across AI ecosystems to standardise communication between AI models and external tools and services.
- Agentic LLM architectures with tool calling are now widely used instead of standalone intent-classification pipelines for complex assistant workflows.
- LangGraph is designed specifically for stateful, multi-step LLM applications where deterministic workflow control, retries and branching are required.
- Visual workflow tools such as LangFlow accelerate prototyping and validation before production deployment.

Recommendation

Retain the existing project objectives but revise implementation tasks from Week 5 onwards to adopt an LLM agent, MCP-based retrieval and LangGraph orchestration. This provides a more scalable architecture while preserving compatibility with the existing backend deliverables.